

STARBUCKS CUSTOMER FEEDBACK INTELLIGENCE & AI-POWERED CUSTOMER RECOVERY

Project Analysis Report

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Data Analysis • Rule-Based Prioritization • Generative AI

1. Executive Summary

This project analyses customer reviews to identify the reviews that require the most attention, understand the main reasons behind negative experiences, and demonstrate how Generative AI can help prepare personalized customer-support responses.

After duplicate removal, 814 reviews remained. Of these, 704 had a rating and were used for rating-based analysis. 549 reviews (78.0%) were classified as critical because they received 1 or 2 stars. Staff / Service was the largest identified complaint area, followed by Pricing / Billing and Order / Payment / App. The priority system identified 64 critical reviews as Urgent and 208 as High priority.

The final stage uses Google Gemini to draft short, personalized apology emails. Human review is required before any outreach, and the optional email workflow is limited to test addresses.

2. Business Problem

The project addresses a practical customer-support problem: a large set of reviews must be converted into a manageable list of critical cases, recurring complaint themes and response drafts. The goal is not to replace human support agents, but to reduce the manual work involved in finding and preparing responses to the most serious cases.

3. Data Overview & Data Quality

Metric	Result
Rows loaded	850
Duplicate review texts	36
Rows after duplicate removal	814
Rated reviews	704
Reviews without rating	110
Average rating	1.87 / 5
Average review length	91 words

Duplicate review texts were removed. Unrated reviews were retained but not assigned an invented rating. Review text was normalized using lowercase conversion, removal of punctuation and special characters, and whitespace cleanup.

4. Exploratory Analysis

Among rated reviews, the average rating is 1.87. The dataset is therefore strongly negative and should be interpreted as a review sample rather than a measure of overall Starbucks customer satisfaction.

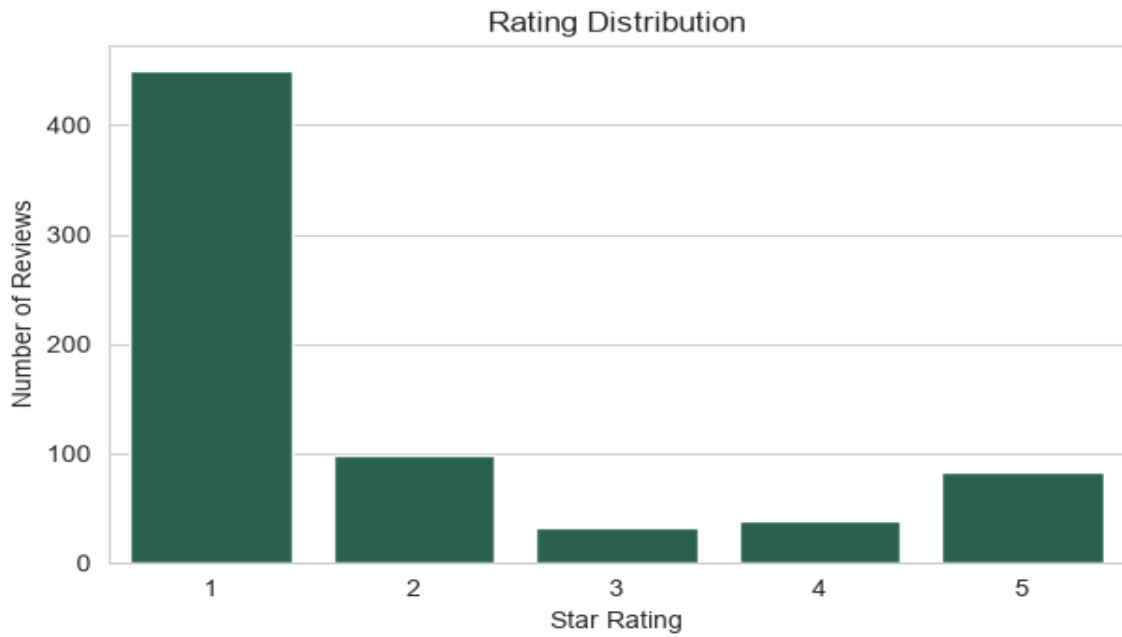


Figure 1. Rating distribution in the analysed review dataset.

5. Critical Review Detection

A review was classified as critical when its rating was 1 or 2 stars.

Review group	Count	Share
Critical (1–2 stars)	549	78.0%
Non-critical (3–5 stars)	155	22.0%

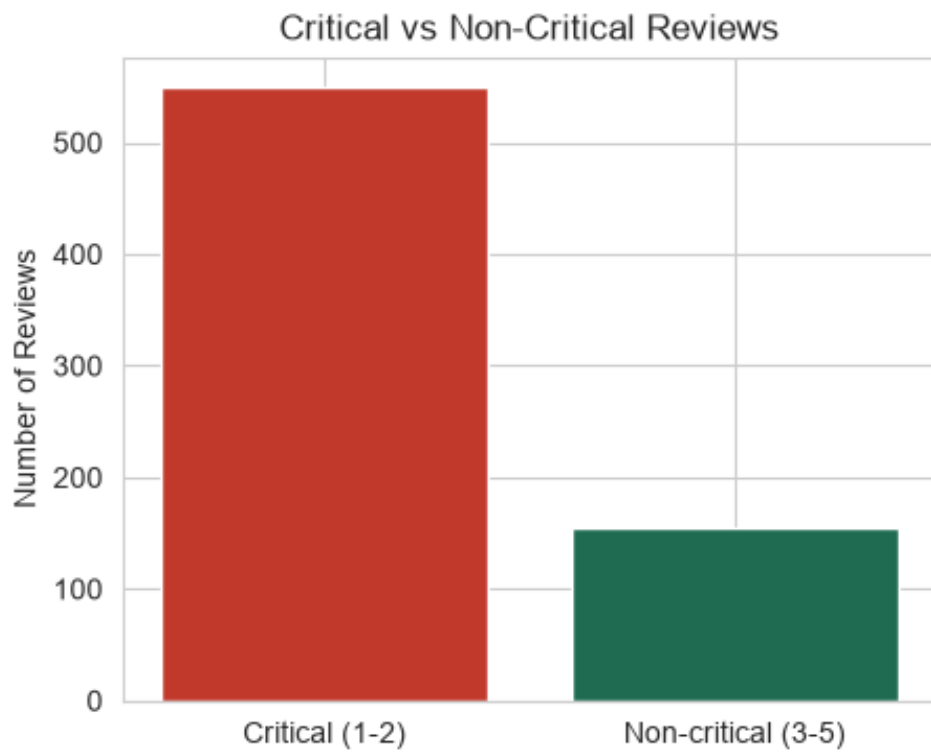


Figure 2. Critical versus non-critical reviews.

6. Complaint Keyword Analysis

The keyword logic counts whether a complaint keyword appears in a review, so repeated use of the same word within one review does not inflate the count.

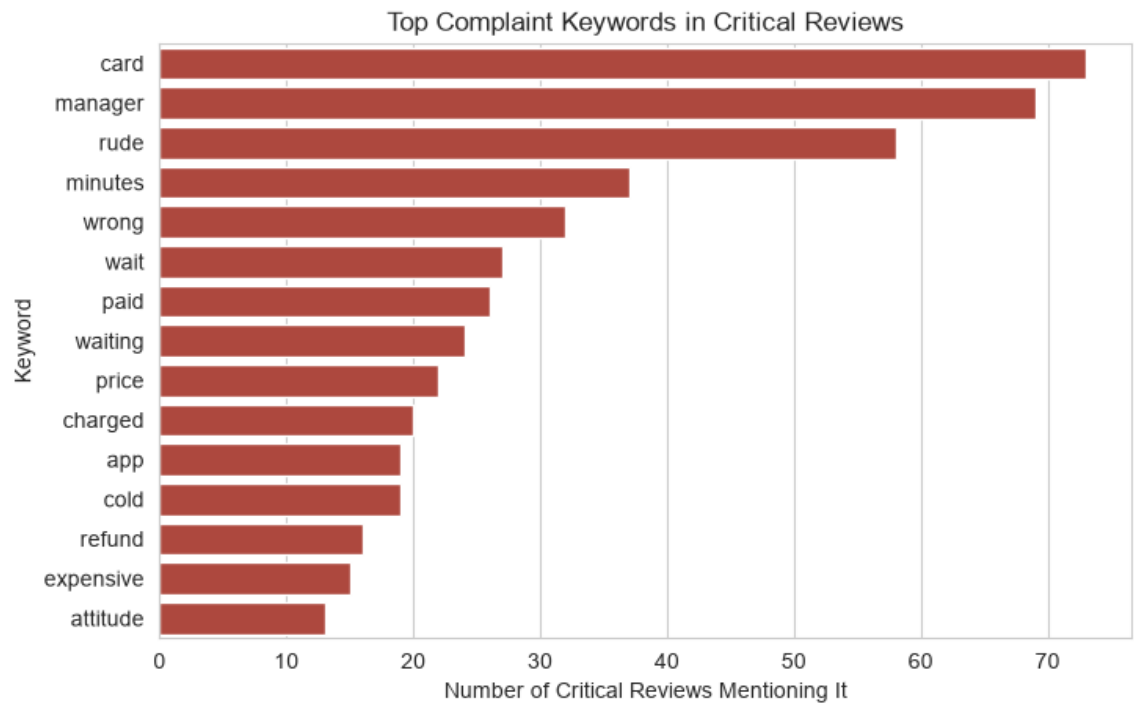


Figure 3. Top complaint keywords in critical reviews.

Card issues appear in 73 critical reviews, manager-related mentions in 69 and rude behaviour in 58. Waiting-time language is also visible, with 'minutes' in 37 critical reviews.

7. Complaint Categorization

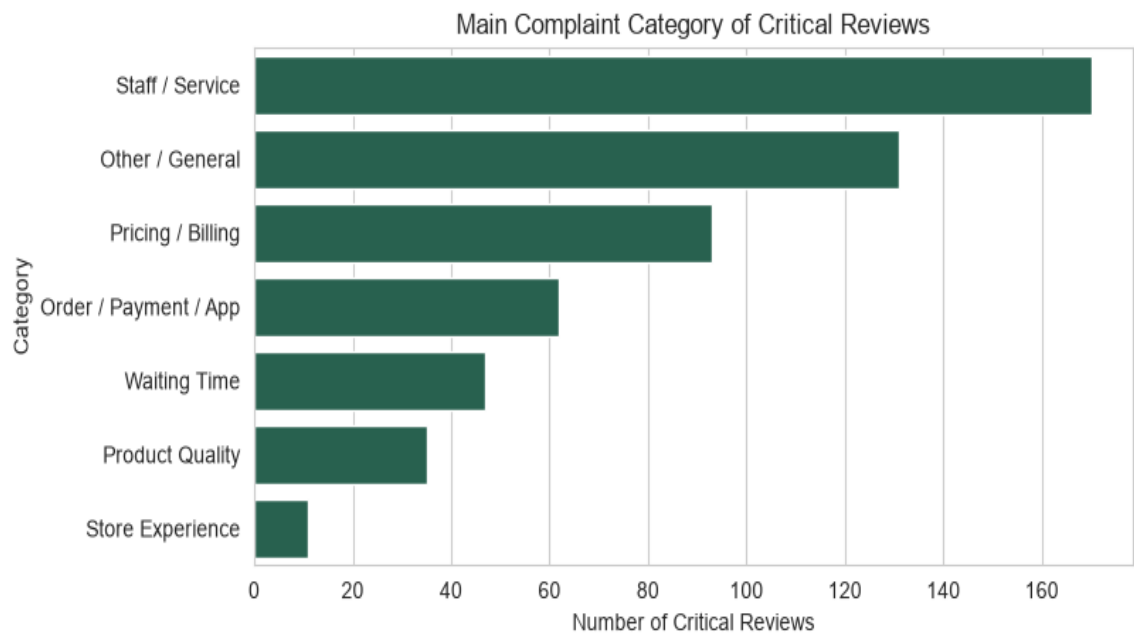


Figure 4. Main complaint categories among critical reviews.

Staff / Service is the largest named complaint group at 170 reviews. Pricing / Billing accounts for 93 and Order / Payment / App for 62. Product Quality is smaller at 35 reviews.

8. Priority Scoring

The priority score combines rating severity, strong red-flag words, review length and multiple complaint categories. The result is a transparent queue rather than a machine-learning prediction.

Priority	Critical reviews
Moderate	277
High	208
Urgent	64

9. Location Analysis

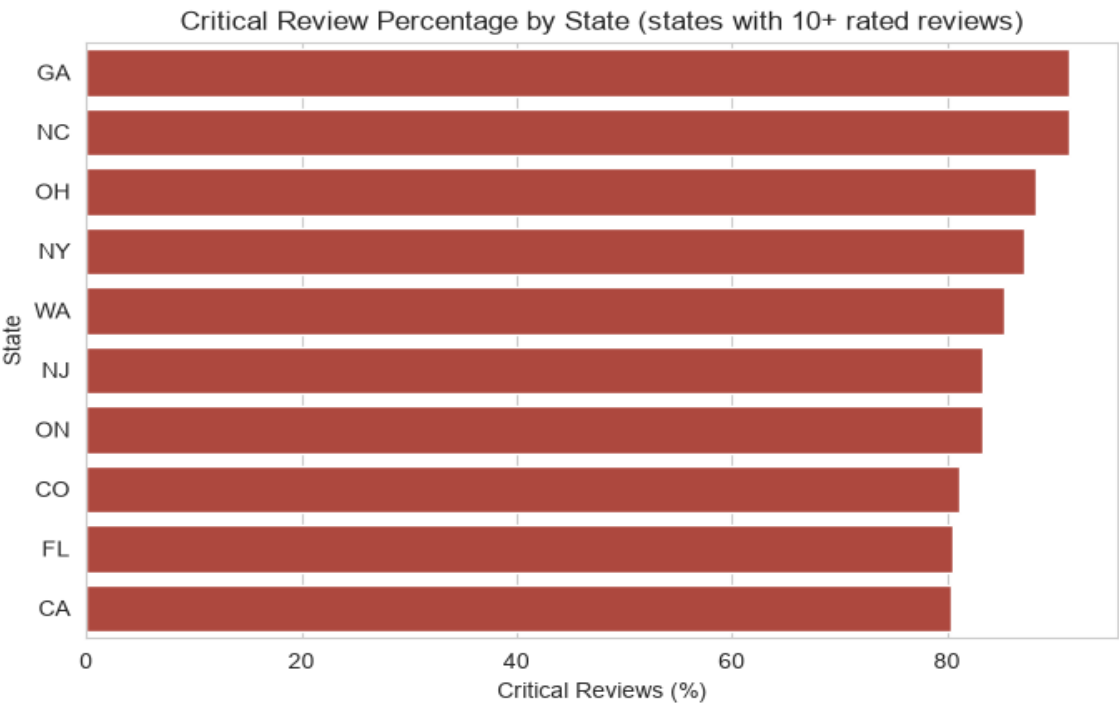


Figure 5. Critical-review percentage by state for states with at least 10 rated reviews.

California has the largest number of critical reviews in the dataset (114). Georgia and North Carolina have the highest critical percentage among states with at least 10 rated reviews (91.3%). These differences are treated cautiously because state sample sizes vary.

10. Top Customer-Recovery Cases

Review ID	Rating	Location	Complaint	Priority
645	1	Houston, TX	Staff / Service	Urgent (8)
694	1	West Allis, WI	Waiting Time	Urgent (8)
563	1	Phoenix, AZ	Staff / Service	Urgent (8)

All three selected cases were rated 1 star and classified as Urgent. They were selected using the highest priority score and review length, giving the GenAI stage detailed cases to work with.

11. Generative AI & Human Review

Google Gemini was used to generate personalized apology-email drafts. The prompt provides the customer name, rating, complaint category, priority and original review, while explicitly preventing invented refunds, compensation, policies or unsupported claims.

The working run successfully generated all three required responses. A human review stage follows the AI generation step so the draft can be checked and edited before any outreach.

The project also includes an optional SMTP test-email workflow. It is disabled by default and is designed to send only approved messages to a test address owned by the user. No real customer outreach is performed because the dataset contains no customer email addresses.

12. Key Insights

- Staff / Service is the largest named complaint area (170 critical reviews).
- Pricing / Billing (93) and Order / Payment / App (62) form a substantial transaction-related complaint group.
- Card issues are the most frequently identified complaint keyword (73 reviews).
- Rude behaviour is mentioned in 58 critical reviews.
- Waiting-time signals are visible in both category and keyword analysis.
- 64 critical reviews are Urgent and 208 are High priority.
- Location differences are exploratory signals rather than strong evidence for broad operational decisions.

13. Recommendations

- Start human review with the Urgent queue.
- Review service and manager-interaction practices where Staff / Service complaints recur.
- Investigate card, refund and billing processes for avoidable customer friction.
- Validate waiting-time complaints with actual store queue and order-preparation data.
- Keep GenAI as an assisted drafting tool with mandatory human approval.
- Collect more location-level observations before making location-based operational changes.
- Track complaint categories over time to measure whether interventions reduce recurring issues.

14. Future Recommendations

- Build a maintainable complaint taxonomy instead of relying only on a hand-written keyword list.
- Add time-series monitoring when reliable review dates are available.
- Connect feedback to store and operational data.
- Add automated quality checks for AI-generated emails.
- Maintain offline response templates for AI outages or quota limitations.
- Create a dashboard for complaint volume, category, priority and response status.
- Measure outcomes using repeat complaints, rating changes and response results.

15. Limitations

- 110 reviews have no rating and are excluded from rating-based analysis.
- Keyword categories may miss context or misclassify some reviews.
- The priority score is a project-level heuristic, not a validated predictive model.
- The dataset is heavily negative and is not representative of overall customer satisfaction.
- State-level comparisons are affected by uneven sample sizes.

- The dataset has no customer email addresses, so real customer outreach cannot be performed from this data alone.

16. Conclusion

The project demonstrates a complete and understandable path from raw customer feedback to assisted customer recovery. It combines practical data cleaning, explainable rule-based filtering, complaint categorization, transparent prioritization, Generative AI drafting and human oversight. The additional test-email workflow makes the prototype closer to a real operational process while keeping actual outreach under control.

The strongest next step would be to connect the review analysis with real operational data and track the same complaint categories over time. That would allow the business to move from identifying complaints to measuring whether corrective actions actually improve customer experience.

17. Technical Stack

Component	Technology
Language	Python
Data processing	Pandas
Text processing	Python string methods + regular expressions
Rule-based analysis	Keyword dictionaries + Counter
Visualisation	Matplotlib / Seaborn
Generative AI	Google Gemini
Email testing	SMTP + Python email library
Environment	Jupyter Notebook